Signals and Systems
SET214
Lecture (12) DT Signals Operations

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Outlines

- Elementary Functions
- Operations
  - Time Shifting
  - Time Compression/Expansion
  - Amplitude Scaling
  - Amplitude Shifting
  - Addition/ Multiplication Signals

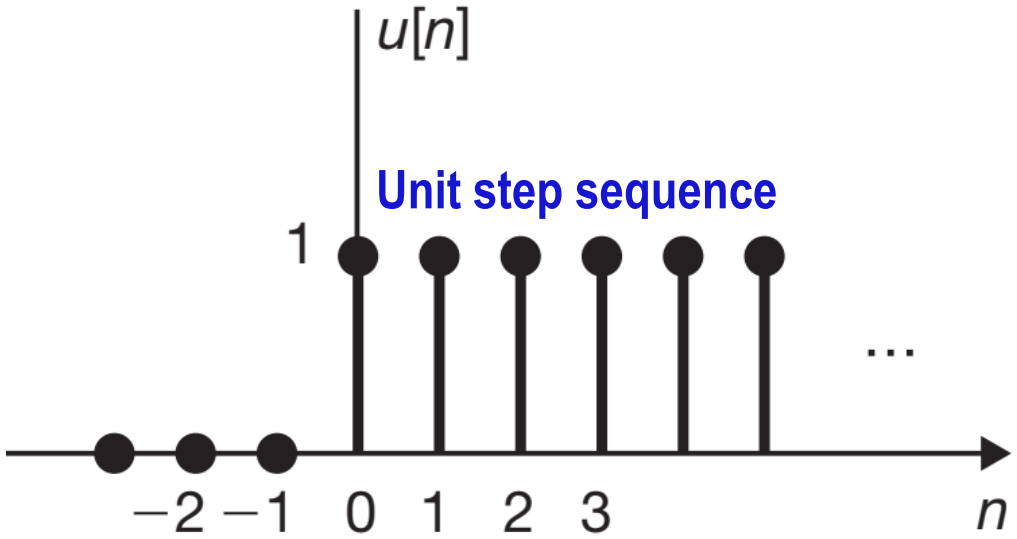
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Elementary Functions (1/3)

- The unit step sequence u[n] is defined as:-

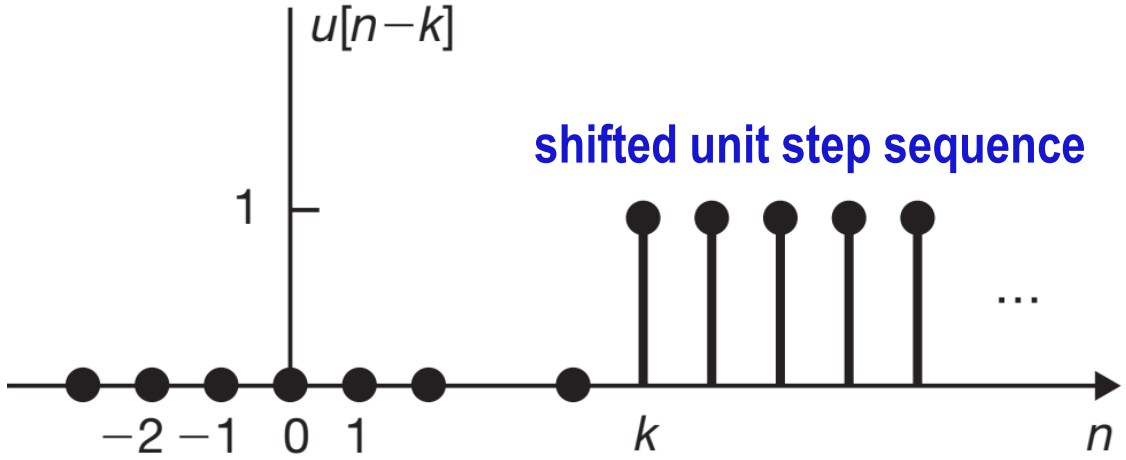
$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



Note that the value of u[n] at n=0 is defined and equals unity

- Similarly, the shifted unit step sequence u[n - k] is defined as:-

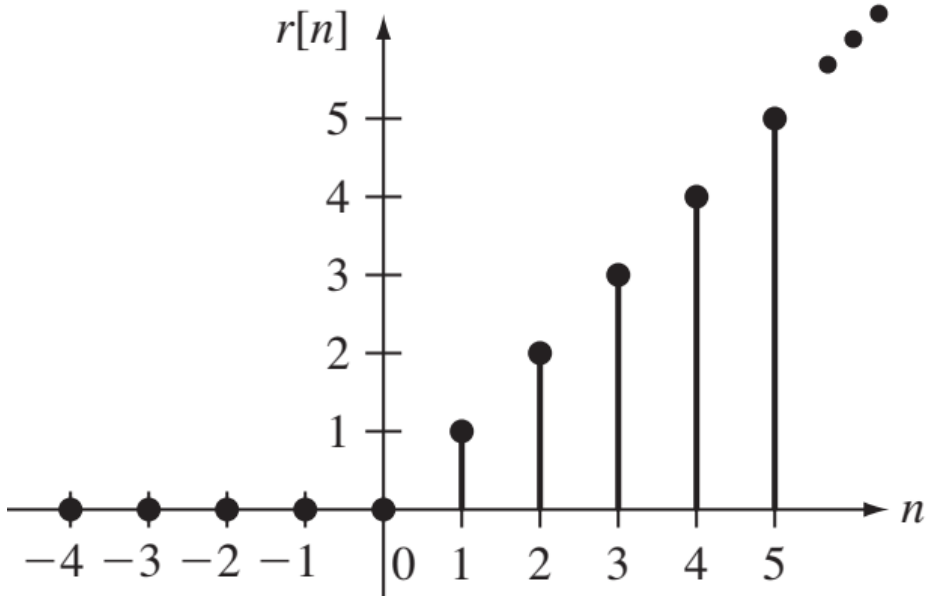
$$u[n-k] = \begin{cases} 1 & n \geq k \\ 0 & n < k \end{cases}$$



Elementary Functions (2/3)

- The discrete-time unit-ramp functions r[n] is defined mathematically by :-

$$r[n] = \begin{cases} n, & n = 0, 1, \ldots \\ 0, & n = -1, -2, \ldots \end{cases}$$

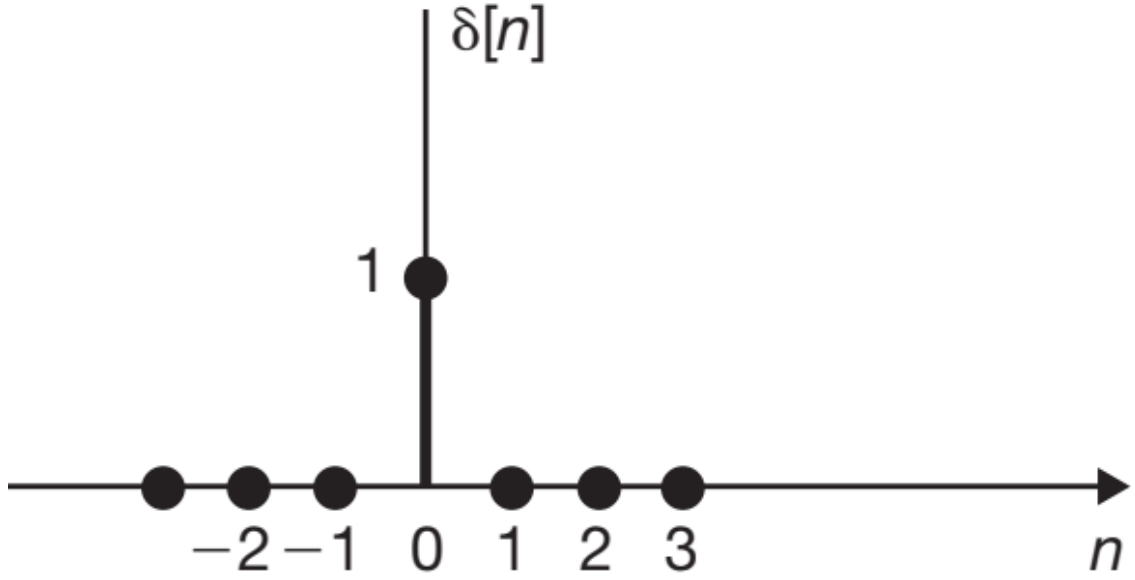


- Note that for the slope of r[n] is 1.
- Thus r[n] has "unit slope," which is the reason r[n] is called the unit-ramp function

Elementary Functions (3/3)

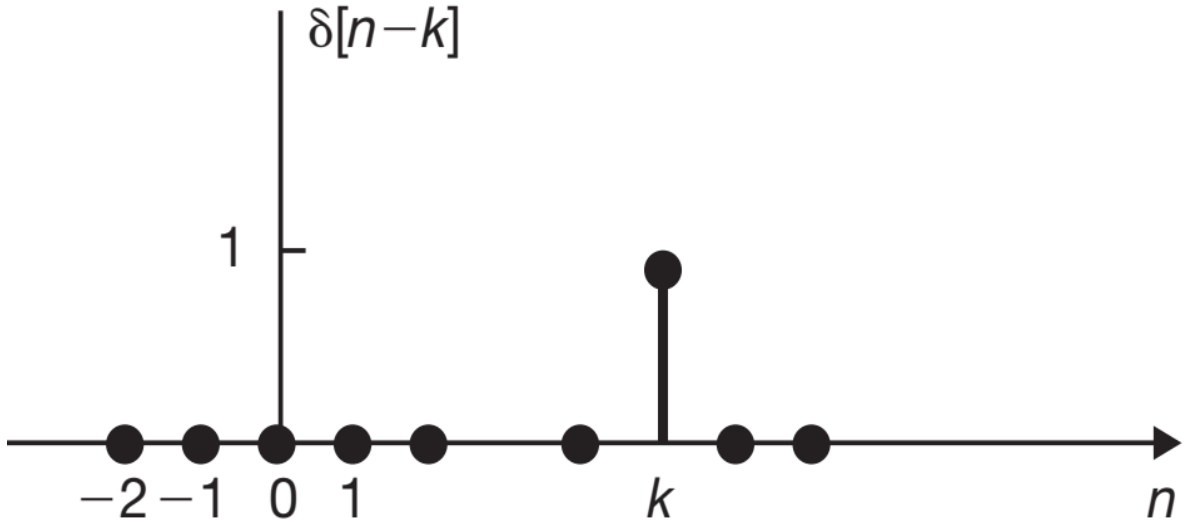
- The unit impulse (or unit sample) sequence δ[n] is defined as:-

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



- Similarly, the shifted unit impulse (or sample) sequence δ[*n* -*k*] is defined as:-

$$\delta[n-k] = \begin{cases} 1 & n = k \\ 0 & n \neq k \end{cases}$$

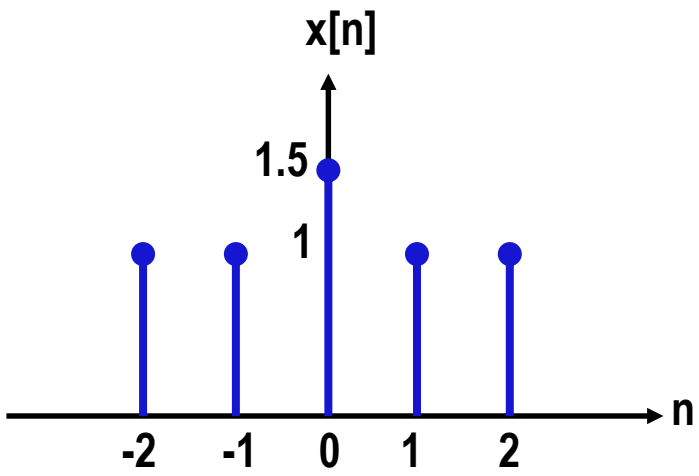


Outlines

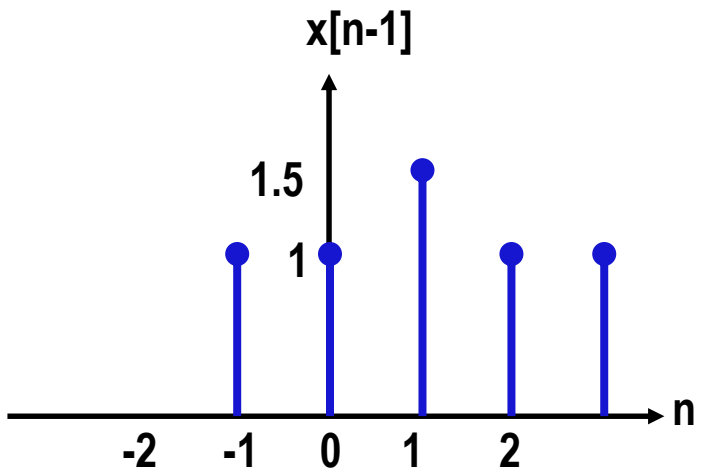
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Time Shifting

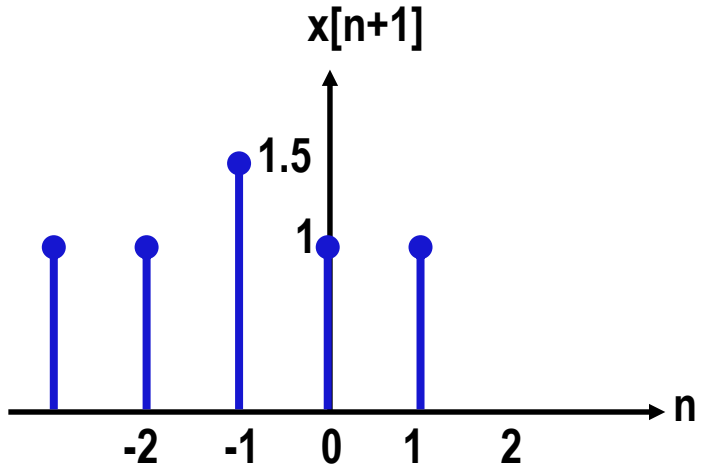
- **Time shifting** (also called **translation**) maps the input function x to the output function y as given by:-
  y[n] = x[n –b], where b is a **real** number
- Such a transformation **shifts** the function **(to the left or right)** along the n axis.
- If b > 0, y is **shifted to the right** by |b|, relative to x (i.e., delayed).
- If b < 0, y is **shifted to the left** by |b|, relative to x (i.e., advanced).



Original Signal



delayed



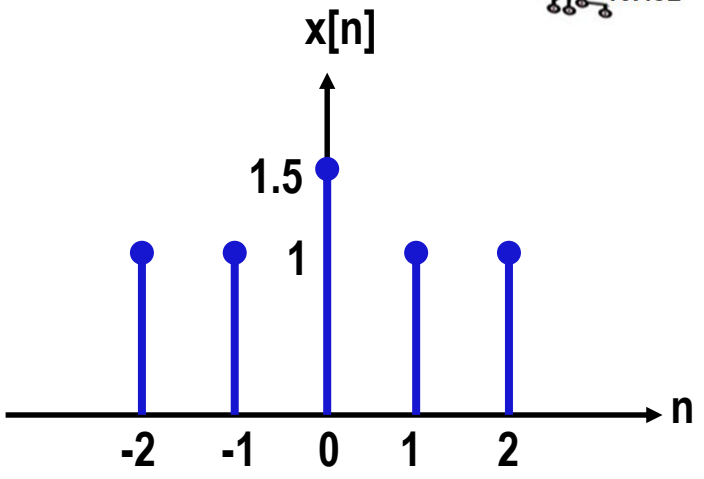
advanced

Outlines

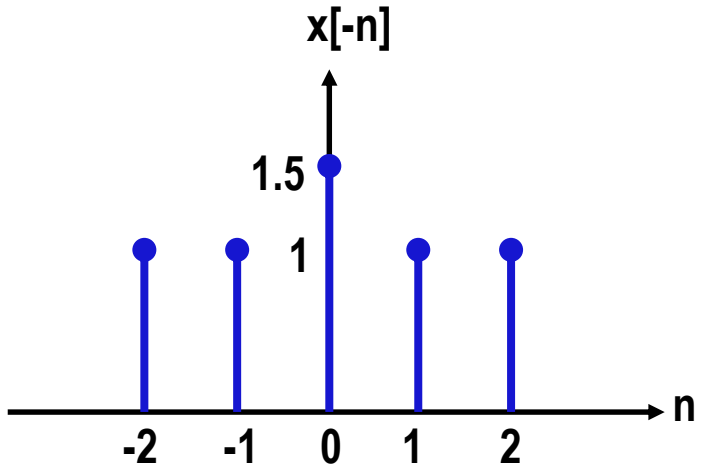
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Time Compression/Expansion

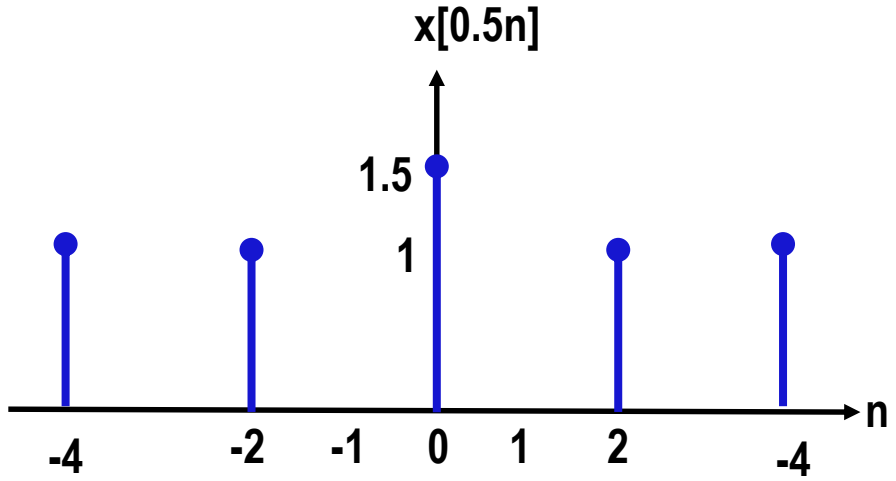
- **Time compression/expansion** (also called **dilation**) maps the input function x to the output function y as given by:-
  y[n] = x[an]; where a is a **strictly positive** real number
- Such a transformation is associated with a **compression/expansion** along the n axis.
- If a > 1, y is **compressed** along the horizontal axis by a factor of a, relative to x.
- If a < 1, y is **expanded** (i.e., stretched) along the horizontal axis by a factor of a , relative to x.
- If a < 0, the function is also time **reversed**



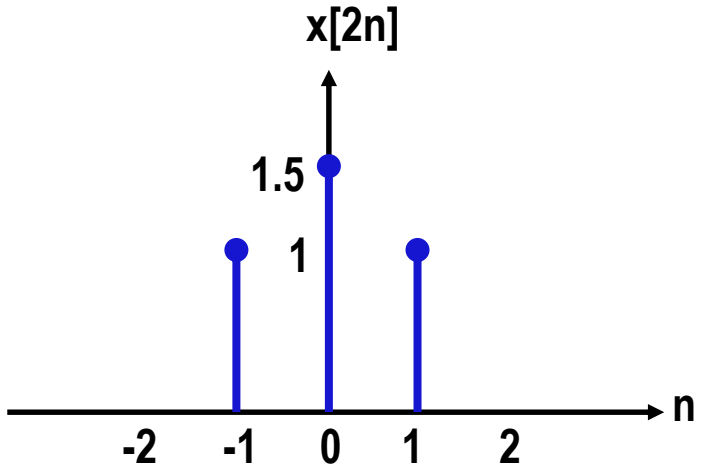
Original Signal



reversed Signal



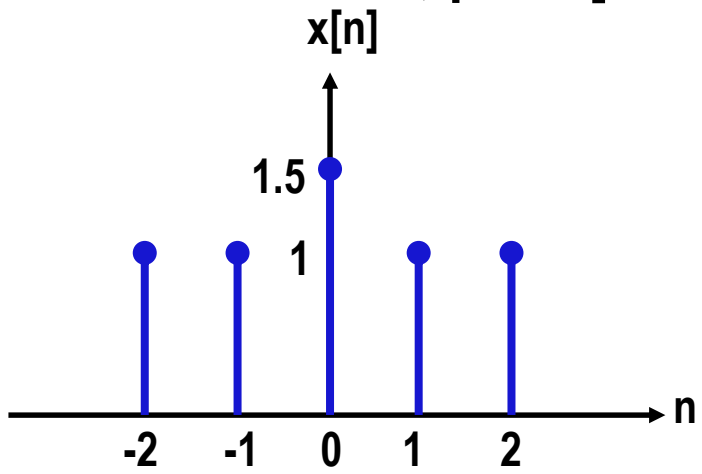
expanded Signal



Compressed

Exercise

Given x[n] as shown below, plot x[2n -1]

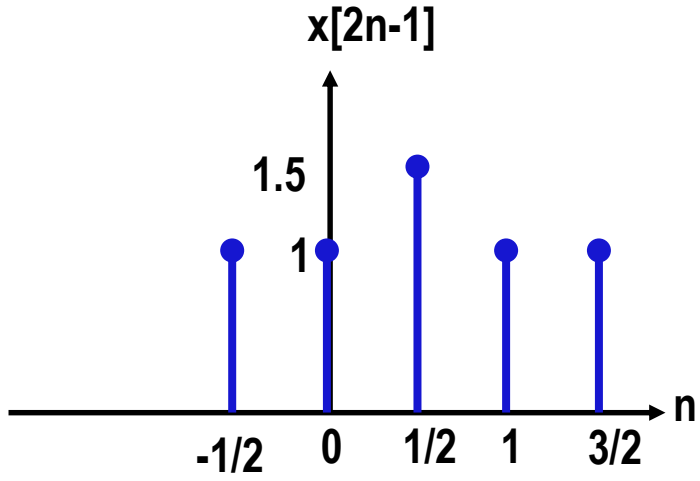


Ans.

❑ Try to build a table as the next

n	X[n]	$n_{new}=(n_{old}+1)/2$	X[2n-1]
-2	1	-1/2	1
-1	1	0	1
0	1.5	1/2	1.5
1	1	1	1
2	1	3/2	1

Notes: Combined Time Scaling and Time Shifting



Outlines

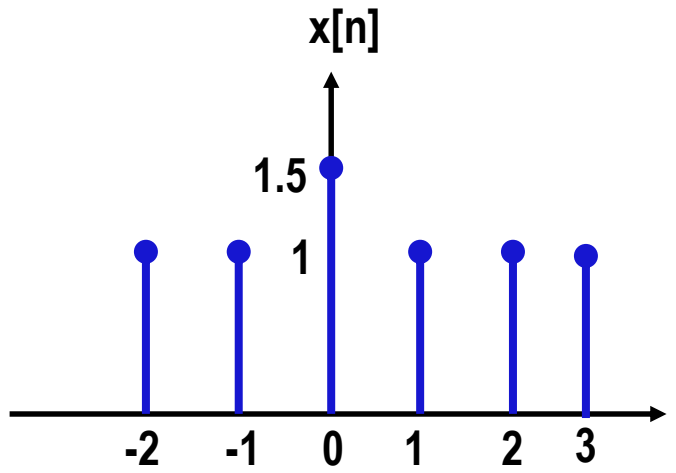
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Amplitude Scaling

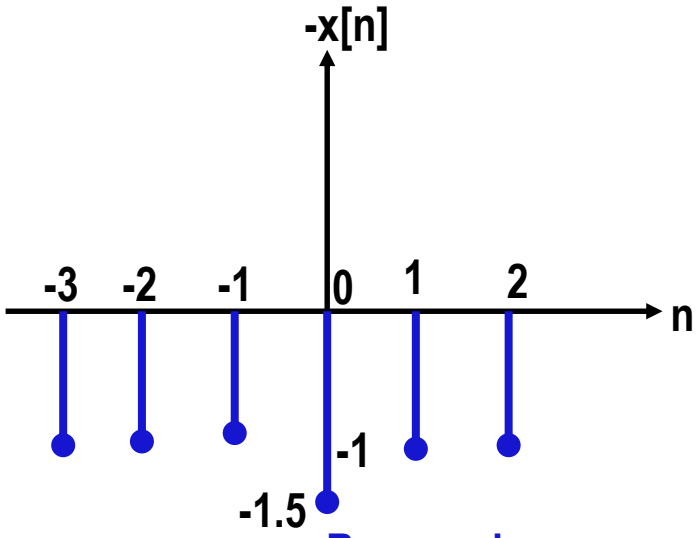
- **Amplitude scaling** maps the input function x to the output function y as given by:-

  y[n] = ax[n], where a is a real number

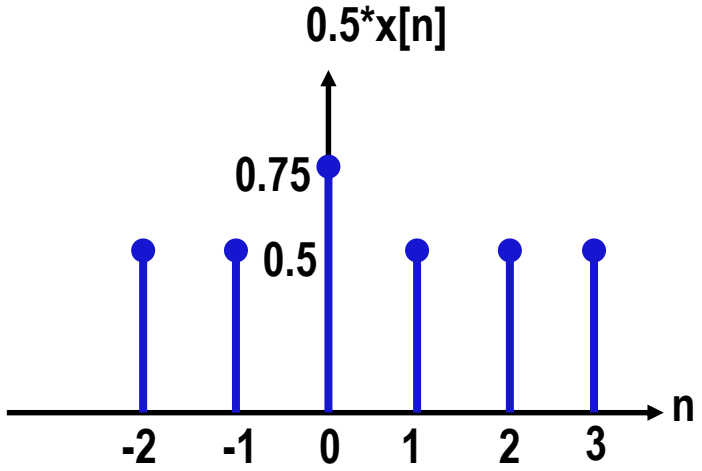
- Geometrically, the output function y is **amplified/attenuated** in amplitude
- If a > 1, y is **amplified** in amplitude by a factor of a
- If a < 1, y is **attenuated in** amplitude by a factor of a
- If a < 0, the function is also amplitude **reversed**



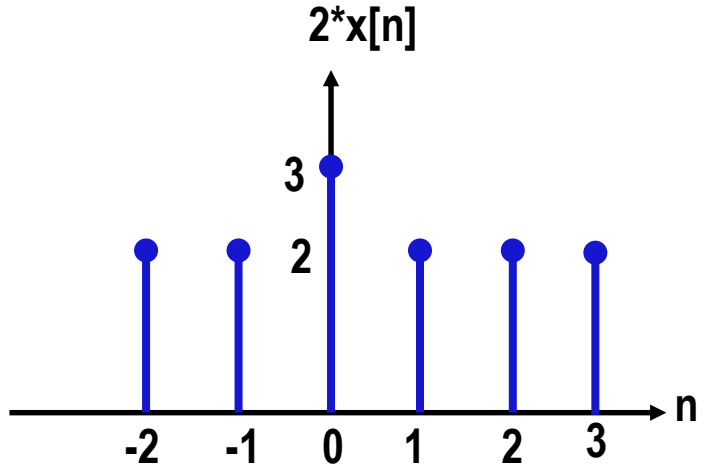
Original Signal



Reversed



Attenuated Signal



Amplified Signal

Outlines

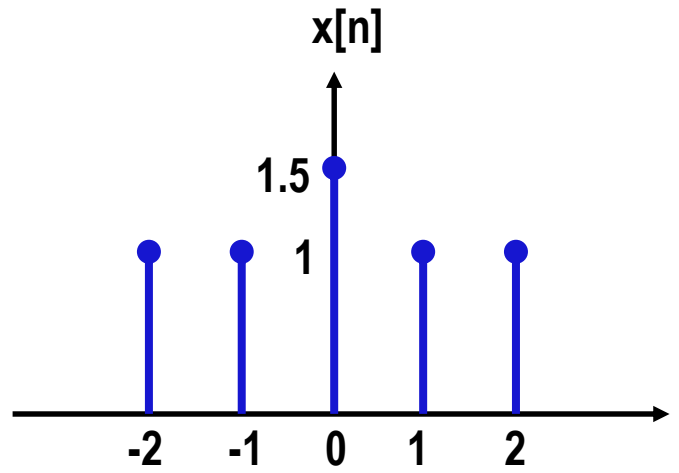
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Amplitude Shifting

- **Amplitude shifting** maps the input function x to the output function y as given by

y[n] = x[n]+ b, where b is a real number

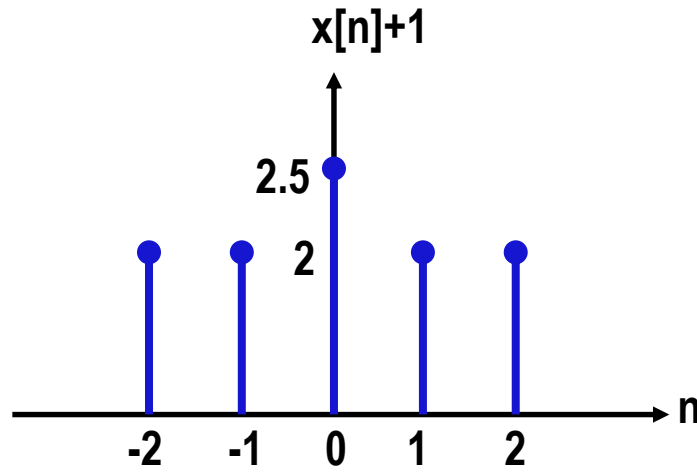
- Geometrically, amplitude **shifting** adds a **vertical displacement** to x



Original Signal

X[n]-1

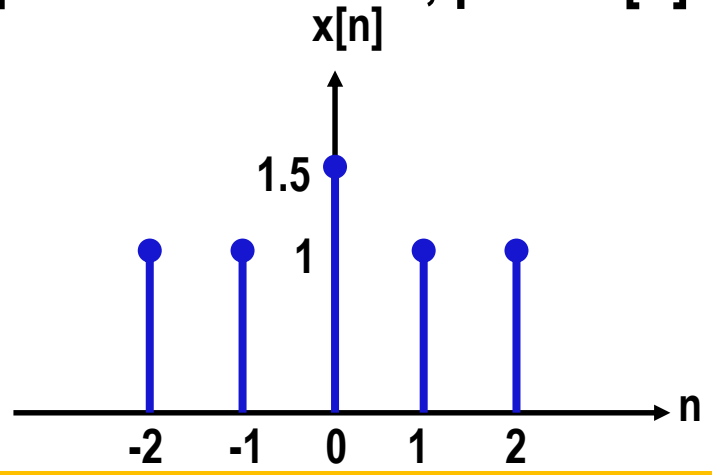
?



Shifted Signal

Exercise

Given x[n] as shown below, plot 2x[n]+1

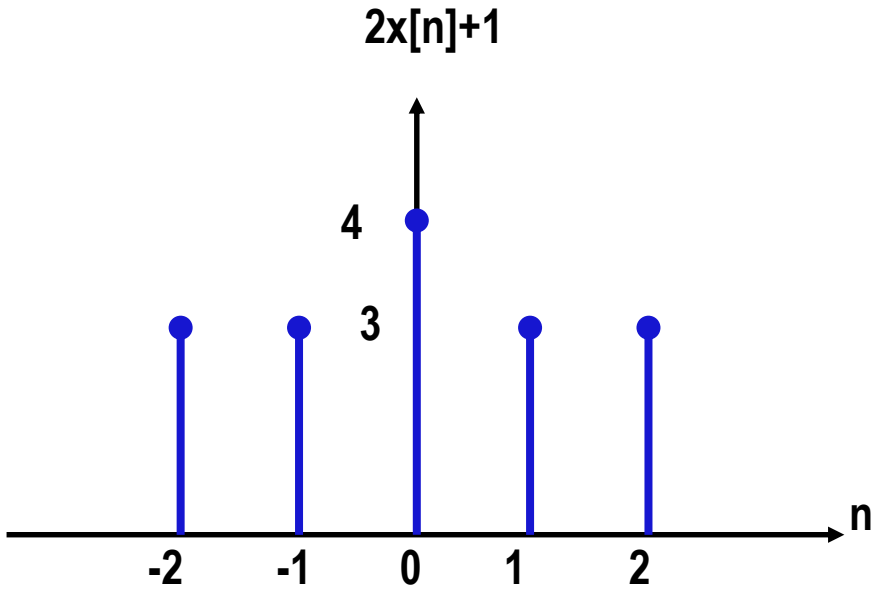


Ans.

❑ Try to build a table as the next

n	X[n]	2x[n]	2x[n]+1
-2	1	2	3
-1	1	2	3
0	1.5	3	4
1	1	2	3
2	1	2	3

Notes: Combined amplitude Scaling and amplitude Shifting

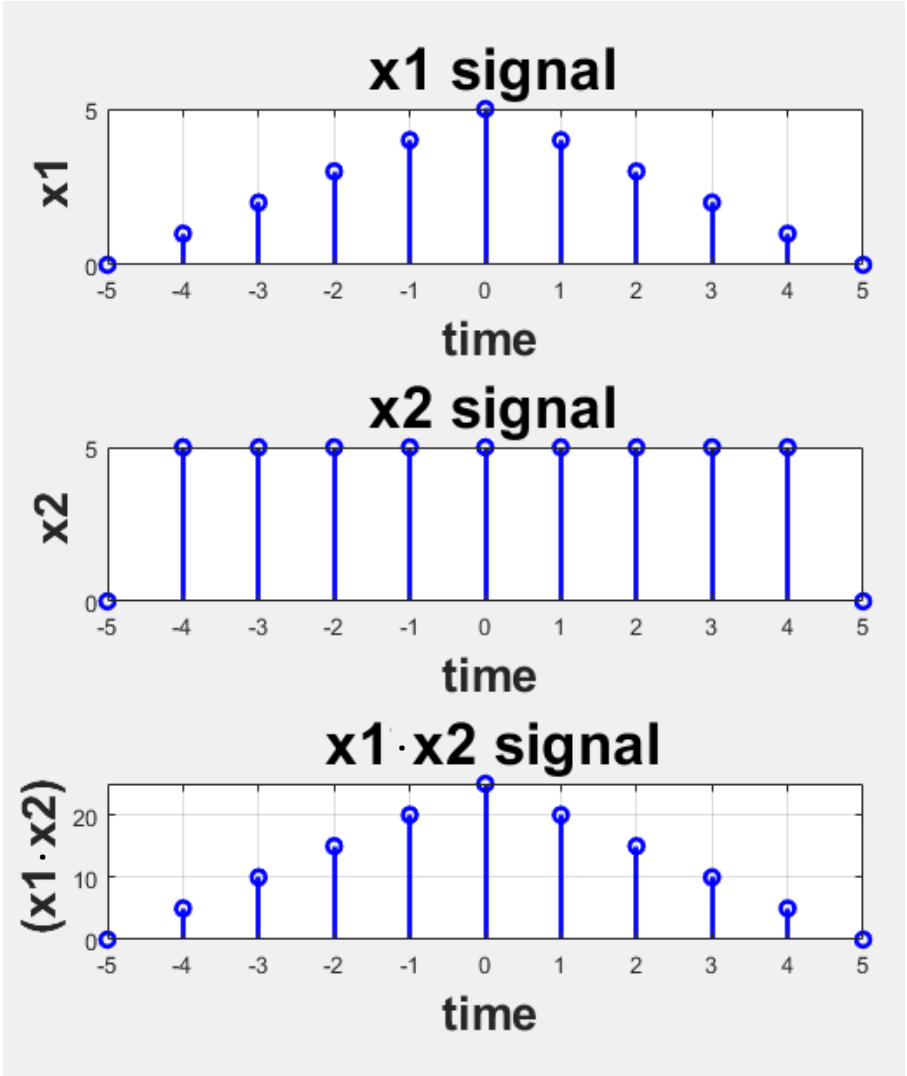


Outlines

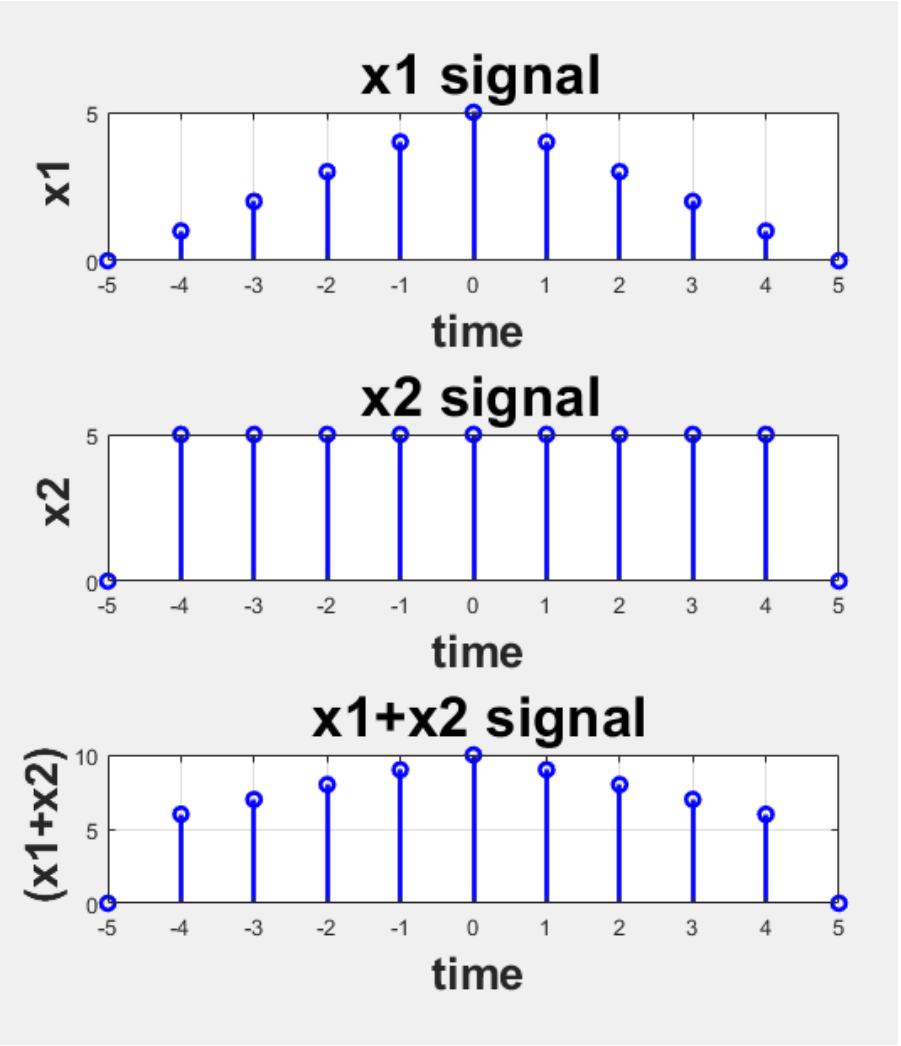
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Addition/multiplication Signals

Two Signals are multiplier.



Two Signals are added.



Thanks!